

CDE4301 Innovation & Design Capstone

Interim Assessment (40%)

This assessment evaluates the **progress** that you have made since the start of your project, and the **quality** of initial work that you have put into realising the intended deliverables of the project. The effort that you put into your project work, as well as the initiative that you show in managing the direction and execution of the project, will also be assessed.

If you are working on a more **design-focused** project, you are expected to perform the following:

- Refine the **problem of interest** and **value proposition** for your project.
- Confirm the **design specifications** that you are required to meet.
- Perform **detail design** of selected concept solution(s).
- Create and test **initial prototypes** of the selected concept solution(s).

If you are working on a more **research-focused** project, you are expected to perform the following:

- Refine the **research gaps** to address.
- Confirm the **hypotheses** that you plan to explore and **scope** of the research.
- Create the **research design matrix** and research protocols (e.g. number of specimens, experimental procedure, etc.) based on the chosen research methodologies.
- Collect and analyse data from **preliminary tests**.

If your project involves a mix of both design and research elements, you are expected to perform the tasks that are appropriate to these elements depending on the context and scope of your work.

In a nutshell, during this interim stage you are expected to deliver a **progress report** for your project which should comprise the following:

- Clear and compelling case about the **novelty** and/or **value** of your project.
- Detailed and viable **plan** with well-articulated steps to accomplish the intended deliverables of the project.
- **Initial outcomes** which demonstrate that there is excellent **potential** of meeting all the intended deliverables of the project.

For this assessment, you are required to give a **presentation** (including prototype demonstration if necessary) of not more than **15 minutes** for **individual** projects or **30 minutes** for **group** projects. The presentation will be held on **Week 12 of Semester 1**.

You are also required to provide an **online report/documentation** (website on GitHub) your project and the work that has been completed in Semester 1. For **individual** projects, the extent of the report/documentation should not exceed **3,000 words** (excluding appendices). For **group** projects, the limit excluding appendices is **4,000 words** for 2 persons; **5,000 words** for 3 persons; **6,000 words** for 4 persons; and **7,000 words** for 5 persons. You should indicate your individual areas of work clearly if you are working in a group project. The **online report/documentation** is due at the end of **Week 13 of Semester 1**.

Your supervisor(s) and examiner will jointly assess the work that you have done based on the content of your presentation and online report/documentation. In addition, your examiner will assess your presentation skills, while your supervisor(s) will assess your effort and initiative throughout Semester 1.

A. Project execution and achievement – graded by SUPERVISOR(S) and EXAMINER

Level of achievement	A1. Objective and scope
<i>Key questions to address</i>	<i>What problem are you trying to address in your project? What are you trying to achieve? How is your project different from existing or past work?</i>
Poor (0 to 2)	☐ Objective of the project is undefined .
Emerging (3 to 5)	☐ Objective and scope of the project are unclear .
Competent (6 to 8)	☐ Objective and scope of the project are clearly defined. ☐ Design specifications are clearly presented and will likely meet the intended deliverables of the project. ☐ Clear articulation of the project's novelty and/or value proposition.
Exceptional (9 to 10)	☐ Objective and scope of the project are clearly defined and feasible . ☐ Design specifications are clearly presented, well justified , and coherent the intended deliverables of the project. ☐ Compelling articulation of the project's novelty and/or value proposition.

Level of achievement	A2. Context of problem
<i>Key questions to address</i>	<i>What is the setting of the problem you are trying to address? Where is your project situated within the bigger scheme of things?</i>
Poor (0 to 2)	☐ Discussion about the context of the problem is almost non-existent .
Emerging (3 to 5)	☐ Discussion about the context of the problem is superficial and/or using information that is mostly irrelevant or outdated .
Competent (6 to 8)	☐ Clear and broad discussion about the context of the problem using information gathered from a range of relevant and up-to-date sources.
Exceptional (9 to 10)	☐ Comprehensive and detailed discussion about the context of the problem using information gathered from an extensive range of relevant, authoritative , and up-to-date sources.

Level of achievement	A3. Methodology / concept development
<i>Key questions to address</i>	<i>How do you intend to achieve the intended deliverables of your project? How do you know that your approach is suitable?</i>
Poor (0 to 2)	☐ Chosen methodologies/concept solutions are unclear .
Emerging (3 to 5)	☐ Chosen methodologies/concept solutions are identified but evaluation/justification and comparison with alternatives are missing .
Competent (6 to 8)	☐ Chosen methodologies/conceptual solutions are clearly identified and evaluated/justified with comparison with alternatives.
Exceptional (9 to 10)	☐ Chosen methodologies/concept solutions and their alternatives are evaluated in a systematic and comprehensive manner.

Level of achievement	A4. Fabrication of specimens/prototypes
<i>Key questions to address</i>	<i>What have you created to achieve the intended deliverables of your project? What are the purposes of these specimens or prototypes? How did you create these specimens or prototypes?</i>
Poor (0 to 2)	☐ Little or no work done to execute the project – prototypes/specimens could have been made, but not carried out.
Emerging (3 to 5)	☐ Prototypes/specimens are selected and/or created but understanding of knowledge, principles, and theories in relevant fields is superficial . Intended purposes of prototypes/specimens may be unclear .
Competent (6 to 8)	☐ Prototypes/specimens are selected and/or created with sound application of knowledge, principles, and theories in relevant fields. Intended purposes of prototypes/specimens are clearly communicated .
Exceptional (9 to 10)	☐ Multiple prototypes/specimens are thoughtfully created with extensive references to and sound application of knowledge/principles/theories in relevant fields. There is clear evidence of fine workmanship in the prototypes/specimens made.

Level of achievement	A5. Testing
<i>Key questions to address</i>	<i>What tests have you performed to investigate the key questions in your project? What are the purposes of these tests? How did you conduct these tests?</i>
Poor (0 to 2)	☐ Little or no work done to execute the project – testing could have been done, but not carried out.
Emerging (3 to 5)	☐ Intended purposes/scope of testing may be unclear , and approaches selected for testing of prototypes/specimens may be unsuitable for the intended purposes.
Competent (6 to 8)	☐ Intended purposes/scope of testing are clearly defined , and testing of prototypes/specimens is conducted satisfactorily using approaches that are appropriate for the intended purposes.
Exceptional (9 to 10)	☐ Intended purposes/scope of testing are clearly defined , and testing of prototypes/specimens is conducted conscientiously using the best possible approaches for the intended purposes.

Level of achievement	A6. Analysis
<i>Key questions to address</i>	<i>What have you found out through the specimens or prototypes which you have created and/or tests which you have conducted in your project? How do you know that your findings are valid?</i>
Poor (0 to 2)	☐ Little or no analysis/explanation of data/observations.
Emerging (3 to 5)	☐ Minimal analysis/explanation of data and observations or using inappropriate methods, techniques, theories, and/or concepts.
Competent (6 to 8)	☐ Analysis/explanation of critical data and observations is performed but use of relevant methods, techniques, theories, and/or concepts to demonstrate or explain key behavior/trends may not be entirely correct .
Exceptional (9 to 10)	☐ In-depth analysis/explanation of data and observations is performed through precise use of appropriate methods, techniques, theories, and/or concepts to demonstrate or explain key behavior/trends.

Level of achievement	A7. Fulfilment of deliverables
<i>Key questions to address</i>	<i>How well has your work addressed the intended deliverables of your project?</i>
Poor (0 to 2)	☐ Little or no outcomes produced.
Emerging (3 to 5)	☐ It is uncertain whether the work done has good potential in meeting the intended deliverables of the project.
Competent (6 to 8)	☐ Able to demonstrate that work done has good potential in meeting most of the intended deliverables of the project.
Exceptional (9 to 10)	☐ Able to demonstrate that work done has excellent potential in meeting all the intended deliverables of the project.

Level of achievement	A8. Awareness of shortcomings
<i>Key questions to address</i>	<i>What are the current shortcomings in your work? How do you plan to resolve them?</i>
Poor (0 to 2)	☐ No awareness of shortcomings of work done.
Emerging (3 to 5)	☐ Some awareness of shortcomings of work done but does not have proposals to resolve these shortcomings.
Competent (6 to 8)	☐ Major shortcomings of work done are identified but proposals to resolve these shortcomings may not be clear or viable .
Exceptional (9 to 10)	☐ Major shortcomings of work done are clearly identified , with sound and well-justified proposals to resolve these shortcomings.

Level of achievement	A9. Project plan
<i>Key questions to address</i>	<i>What do you intend to do in order to ensure that you would be able to complete your project successfully?</i>
Poor (0 to 2)	☐ Plans for completing the project are non-existent .
Emerging (3 to 5)	☐ Has a simple plan for completing the project, but milestones are unclear .
Competent (6 to 8)	☐ Able to propose reasonable plans for completing the project with some rough milestones to be reached.
Exceptional (9 to 10)	☐ Able to propose detailed and viable plans for completing the project, with well-articulated steps and clearly identified milestones to be reached.

B. Presentation – graded by EXAMINER only

Level of achievement	B1. Quality of presentation material
Poor (0 to 2)	☐ Presentation material is missing many key parts, inconsistent , disorganized, and/or contains many major errors . ☐ Little or no use of illustrations to communicate key ideas/information.
Emerging (3 to 5)	☐ Presentation material lacks visual appeal and contains many minor errors and inconsistencies that are distracting and/or may cause some confusion. ☐ Illustrations are ill-chosen (i.e. not fitting for their intended purposes) and/or lack clear annotations.
Competent (6 to 8)	☐ Presentation material is visually pleasing and well-organized but may contain minor errors and inconsistencies that are not distracting or confusing. ☐ Good use of illustrations to communicate key ideas/information but could be more clearly annotated .
Exceptional (9 to 10)	☐ Presentation material has excellent visual appeal , well-organized and virtually error-free , and looks professionally made . ☐ Extensive use of properly annotated and easily understood illustrations to communicate key ideas/information.

Level of achievement	B2. Delivery and presentation skills
Poor (0 to 2)	☐ Presentation is barely comprehensible and throughout. ☐ Presentation is awkward throughout. There is little evidence of practice/preparation.
Emerging (3 to 5)	☐ Transition/sequence of ideas lacks clarity and/or is abrupt at times, and/or work done tends to be presented in chronological rather than logical sequence. ☐ Some evidence of practice/preparation, but time management needs to be improved and/or pace of delivery is too slow or too fast.
Competent (6 to 8)	☐ Clear transition/sequence of ideas but logical flow can be improved and/or presentation can be more concise . ☐ Good time management and pace of delivery. Clear evidence of practice/preparation but can be more confident.
Exceptional (9 to 10)	☐ Clear, comprehensive, concise , and compelling presentation with excellent logical flow . ☐ Precise time management and pace of delivery. Confident presentation with clear evidence of practice/preparation

Level of achievement	B3. Response to questions/comments
Poor (0 to 2)	☐ Unable to respond to questions/comments. Little evidence of practice/preparation.
Emerging (3 to 5)	☐ Needs repeated prompting to respond to questions/comments. ☐ Responses to questions/comments are superficial and suggest lack of understanding about the subject matter.
Competent (6 to 8)	☐ Responses to questions/comments are generally accurate and may sometimes lack clarity and/or could be more concise.
Exceptional (9 to 10)	☐ Responses to questions/comments are thoughtful, accurate , and concise . Clear evidence of practice/preparation.

C. Effort and Initiative – graded by SUPERVISOR(S) only

Level of achievement	C1. Motivation
Poor (0 to 2)	☐ Unmotivated and demonstrates minimal attempt to achieve required tasks. ☐ Frequently makes excuses for not carrying out work.
Emerging (3 to 5)	☐ Attempts to achieve required tasks but needs to be prompted frequently. ☐ Tends to stop working when difficulties arise
Competent (6 to 8)	☐ Well-motivated and demonstrates consistent attempts to execute required tasks. ☐ Attempts to continue working when difficulties arise.
Exceptional (9 to 10)	☐ Highly motivated and always exerts maximal attempts to execute required tasks. ☐ Always perseveres/persistent in tackling problems encountered even when solutions may not be immediately obvious.

Level of achievement	C2. Attitude towards project meetings
Poor (0 to 2)	☐ Frequently skips or late for meetings.
Emerging (3 to 5)	☐ Punctual for meetings but is unprepared for and/or seldom contributes during meetings.
Competent (6 to 8)	☐ Mostly punctual and well-prepared as well as contributes actively during meetings.
Exceptional (9 to 10)	☐ Always punctual and well-prepared as well as contributes actively and thoughtfully during meetings.

Level of achievement	C3. Independent inquiry
Poor (0 to 2)	☐ Demonstrates no interest in exploring relevant questions for further inquiry unless instructed. ☐ Persistent guidance or help from supervisors is given but not followed .
Emerging (3 to 5)	☐ Demonstrates some interest in exploring relevant questions but may not act on tasks for further inquiry. ☐ Guidance or help from supervisors is frequently required but sometimes not followed.
Competent (6 to 8)	☐ Generally able to generate relevant questions and execute tasks for further inquiry. ☐ Attempts to learn and overcome problems independently but sometimes require guidance or help from supervisors
Exceptional (9 to 10)	☐ Constantly generates relevant questions and executes tasks for further inquiry. ☐ Able to learn and overcome problems with minimal guidance or help from supervisors.

Level of achievement	C4. Ownership and resourcefulness
Poor (0 to 2)	☐ Displays zero ownership of the project.
Emerging (3 to 5)	☐ Ownership of the project mostly by supervisors . ☐ Frequently waits for supervisors to provide resources for the project.
Competent (6 to 8)	☐ Frequently attempts to self-manage the direction and execution of the project. ☐ Actively proposes resources required for the project to supervisors.
Exceptional (9 to 10)	☐ Displays clear ownership and leadership in self-managing the direction and execution of the project. ☐ Actively proposes and secures resources for the project on his/her own.